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In [1]: #import statements  
import pandas as pd  
import matplotlib.pyplot as plt  
%matplotlib inline  
from pandas.plotting import scatter_matrix
```

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In [14]: # Specify the fields with their names  
fields = ['preg', 'plas', 'pres', 'skin', 'test', 'mass', 'pedi', 'age', 'cl  
# load the pima_diabetes dataset  
df = pd.read_csv("Assignment_4pima_diabetes.csv", names= fields)  
# Printing first 45 records  
df.head(45)
```

Out[14]:

	preg	plas	pres	skin	test	mass	pedi	age	class
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1
5	5	116	74	0	0	25.6	0.201	30	0
6	3	78	50	32	88	31.0	0.248	26	1
7	10	115	0	0	0	35.3	0.134	29	0
8	2	197	70	45	543	30.5	0.158	53	1
9	8	125	96	0	0	0.0	0.232	54	1
10	4	110	92	0	0	37.6	0.191	30	0
11	10	168	74	0	0	38.0	0.537	34	1
12	10	139	80	0	0	27.1	1.441	57	0
13	1	189	60	23	846	30.1	0.398	59	1
14	5	166	72	19	175	25.8	0.587	51	1
15	7	100	0	0	0	30.0	0.484	32	1
16	0	118	84	47	230	45.8	0.551	31	1
17	7	107	74	0	0	29.6	0.254	31	1
18	1	103	30	38	83	43.3	0.183	33	0
19	1	115	70	30	96	34.6	0.529	32	1
20	3	126	88	41	235	39.3	0.704	27	0
21	8	99	84	0	0	35.4	0.388	50	0
22	7	196	90	0	0	39.8	0.451	41	1
23	9	119	80	35	0	29.0	0.263	29	1
24	11	143	94	33	146	36.6	0.254	51	1
25	10	125	70	26	115	31.1	0.205	41	1
26	7	147	76	0	0	39.4	0.257	43	1
27	1	97	66	15	140	23.2	0.487	22	0
28	13	145	82	19	110	22.2	0.245	57	0
29	5	117	92	0	0	34.1	0.337	38	0
30	5	109	75	26	0	36.0	0.546	60	0
31	3	158	76	36	245	31.6	0.851	28	1

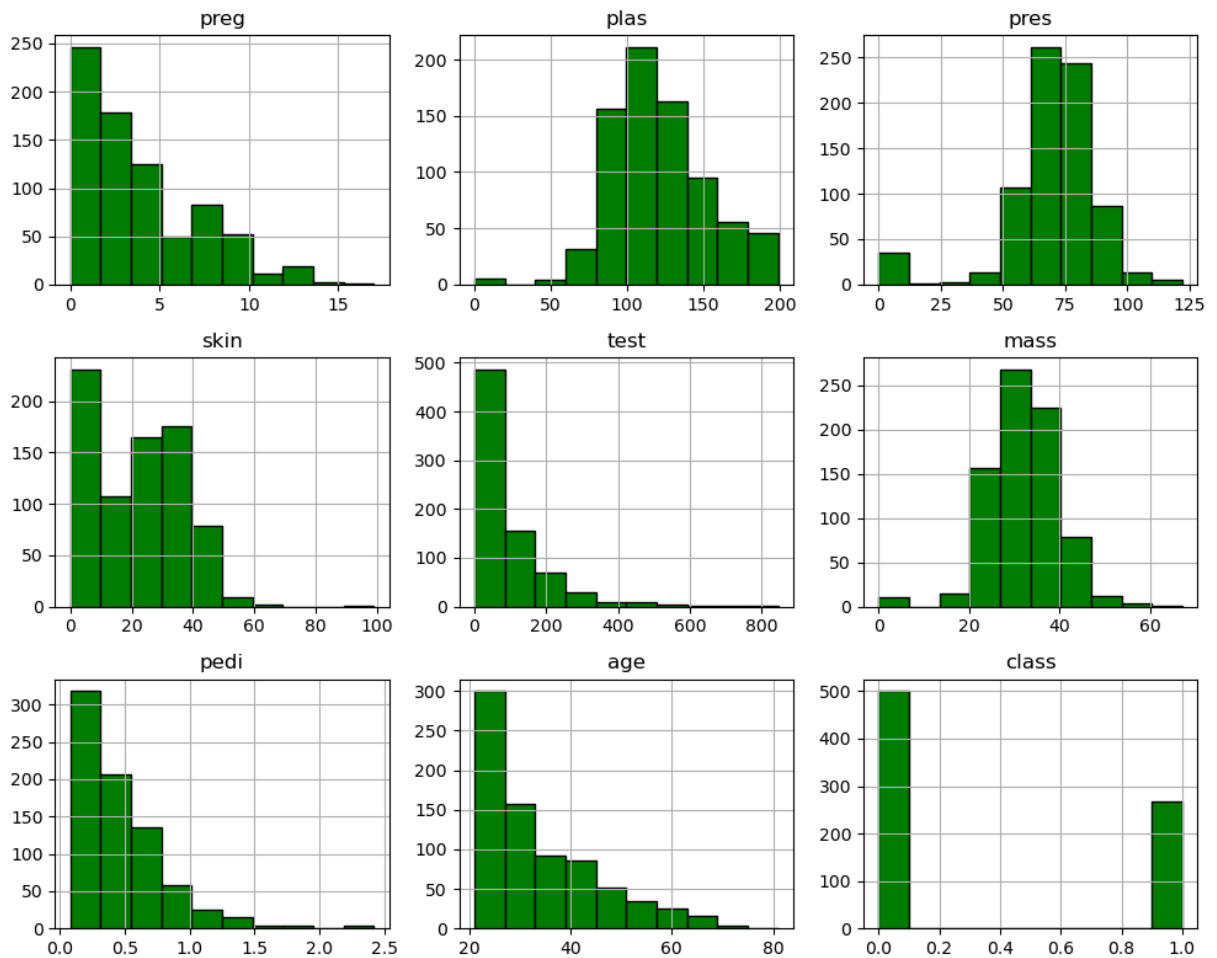
	preg	plas	pres	skin	test	mass	pedi	age	class
32	3	88	58	11	54	24.8	0.267	22	0
33	6	92	92	0	0	19.9	0.188	28	0
34	10	122	78	31	0	27.6	0.512	45	0
35	4	103	60	33	192	24.0	0.966	33	0
36	11	138	76	0	0	33.2	0.420	35	0
37	9	102	76	37	0	32.9	0.665	46	1
38	2	90	68	42	0	38.2	0.503	27	1
39	4	111	72	47	207	37.1	1.390	56	1
40	3	180	64	25	70	34.0	0.271	26	0
41	7	133	84	0	0	40.2	0.696	37	0
42	7	106	92	18	0	22.7	0.235	48	0
43	9	171	110	24	240	45.4	0.721	54	1
44	7	159	64	0	0	27.4	0.294	40	0

```
In [17]: # Printing last 25 records  
df.tail(25)
```

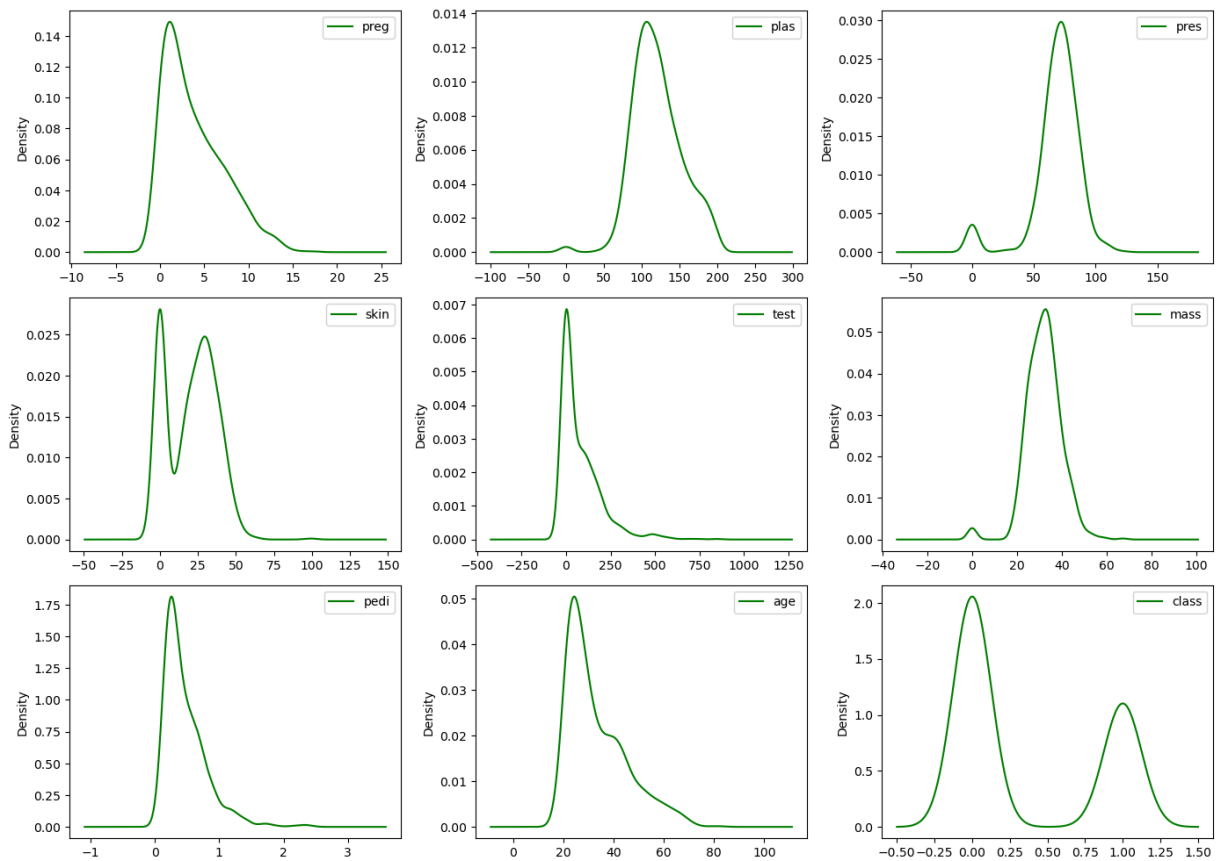
Out [17]:

	preg	plas	pres	skin	test	mass	pedi	age	class
743	9	140	94	0	0	32.7	0.734	45	1
744	13	153	88	37	140	40.6	1.174	39	0
745	12	100	84	33	105	30.0	0.488	46	0
746	1	147	94	41	0	49.3	0.358	27	1
747	1	81	74	41	57	46.3	1.096	32	0
748	3	187	70	22	200	36.4	0.408	36	1
749	6	162	62	0	0	24.3	0.178	50	1
750	4	136	70	0	0	31.2	1.182	22	1
751	1	121	78	39	74	39.0	0.261	28	0
752	3	108	62	24	0	26.0	0.223	25	0
753	0	181	88	44	510	43.3	0.222	26	1
754	8	154	78	32	0	32.4	0.443	45	1
755	1	128	88	39	110	36.5	1.057	37	1
756	7	137	90	41	0	32.0	0.391	39	0
757	0	123	72	0	0	36.3	0.258	52	1
758	1	106	76	0	0	37.5	0.197	26	0
759	6	190	92	0	0	35.5	0.278	66	1
760	2	88	58	26	16	28.4	0.766	22	0
761	9	170	74	31	0	44.0	0.403	43	1
762	9	89	62	0	0	22.5	0.142	33	0
763	10	101	76	48	180	32.9	0.171	63	0
764	2	122	70	27	0	36.8	0.340	27	0
765	5	121	72	23	112	26.2	0.245	30	0
766	1	126	60	0	0	30.1	0.349	47	1
767	1	93	70	31	0	30.4	0.315	23	0

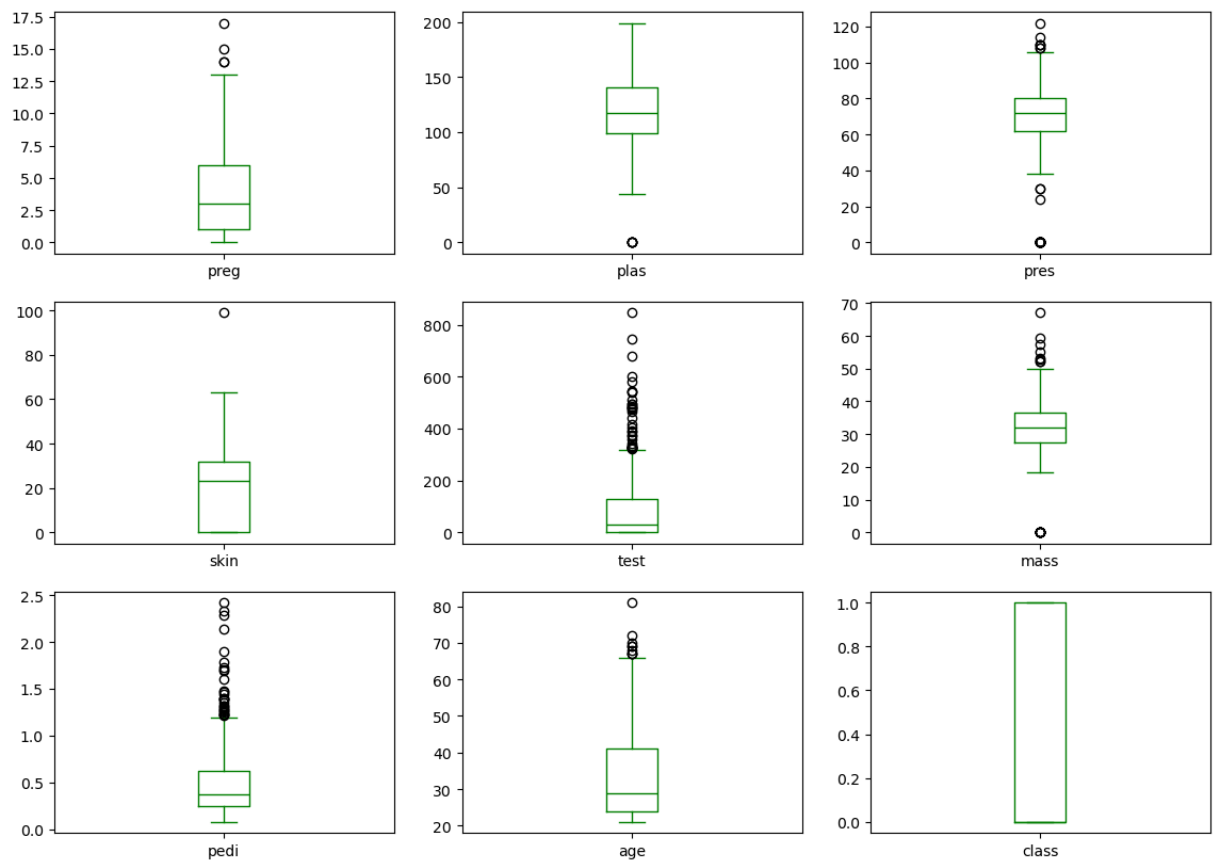
```
In [74]: # creating a histogram plot for all numeric variables in the dataset
df.hist(figsize=(10,8),color='green', edgecolor="black")
plt.tight_layout()
plt.show()
```



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In [81]: # Density Plot for all numeric variables in the dataset
df.plot(kind='density',subplots=True,layout=(3, 3),figsize=(14, 10),color='g')
plt.tight_layout()
plt.show()
```

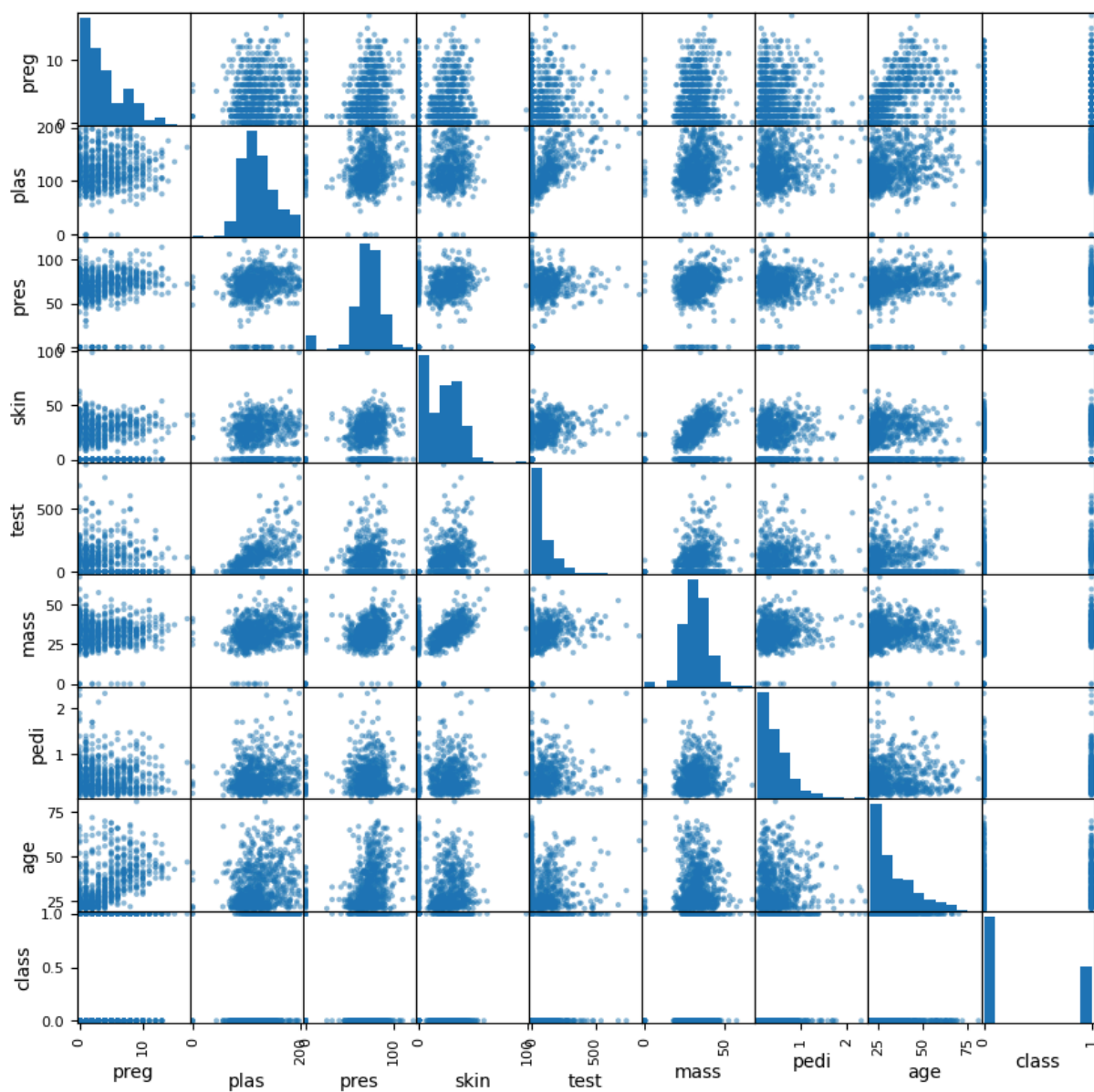


```
In [64]: # Box Plot for all numeric variables in the dataset
df.plot(kind='box',subplots=True,layout=(3, 3),figsize=(14, 10),color='green',
plt.show())
```



```
In [79]: # Creating a scatter matrix plot for numeric variables to visualize pairwise
scatter_matrix(df, alpha=0.5, figsize=(10,10))
plt.suptitle('Scatter Matrix plot for numeric variables')
plt.show()
```

Scatter Matrix plot for numeric variables



In []:

In []:

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